



Datos, evidencia, decisiones:
generando valor para la gestión
y las políticas sanitarias

Sevilla, 17 al 19 de junio de 2026



UNIVERSIDAD DE
COSTA RICA



EEC Escuela de
Economía

Cancer Mortality Atlas of Costa Rica, 2000–2023: A Bayesian Spatial Estimation Approach

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Motivation: the small-area estimation problem

Problem:

$$\text{Crude SMR} = O_i / E_i \times 100$$

- O_i : observed deaths in county i ; E_i : expected deaths under national risk.
- In small-population counties, $E_i \approx 0$:
 - $O_i = 0 \Rightarrow \text{SMR} = 0$ (false protection)
 - $O_i = 1 \Rightarrow \text{SMR} \rightarrow \infty$ (false alarm)
- **Sampling noise $\neq$ real risk pattern.**

Solution: Methodological Contribution

Bayesian spatial smoothing (BYM, 1991):

- Neighbouring counties share information through a spatial prior.
- MCMC produces the full posterior distribution of risk.
- Smoothed SMRs with explicit uncertainty quantification.

Geographic scope:

- 82 counties of Costa Rica, 2000–2023
- 18 tumour sites (ICD-10 classification), year, sex, age, population
- Source: INEC Costa Rica



The Besag, York and Mollie model (BYM, 1991)

Observation Model

Counts are Poisson: $O_i \mid R_i \sim \text{Poisson}(E_i R_i)$

Log-risk decomposition:

$$\log(R_i) = \underbrace{\mu}_{\text{global}} + \underbrace{\alpha_i}_{\text{spatial}} + \underbrace{\beta_i}_{\text{heterog.}}$$

- $\mu = \mathbf{X}_i \boldsymbol{\theta}$: global component (age, sex); diffuse prior $\boldsymbol{\theta} \sim \mathcal{N}(-\infty, +\infty)$.
- α_i : structured spatial effect (ICAR).
- β_i : unstructured (heterogeneous) effect.

Prior Structure

Structured spatial prior (ICAR):

$$\alpha_i \mid \boldsymbol{\alpha}_{-i}, \sigma^2 \sim \mathcal{N}\left(\bar{\alpha}_i, \frac{\sigma^2}{n_i}\right)$$

- $\bar{\alpha}_i$ = mean of α_j over neighbours of i .
- n_i = number of neighbours of county i .

Unstructured heterogeneity: $\beta_i \mid \tau^2 \sim \mathcal{N}(0, \tau^2)$

Hyperpriors: $\sigma^2, \tau^2 \sim \Gamma^{-1}(1, 0.01)$ (vague, non-informative)



Joint posterior distribution and MCMC rationale

$$p(\alpha, \beta, \theta, \sigma^2, \tau^2 \mid \mathbf{O}) \propto \underbrace{\prod_{i=1}^n e^{-E_i e^{\log R_i}} (E_i e^{\log R_i})^{O_i}}_{\text{Poisson likelihood}} \times \underbrace{(\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i \sim j} (\alpha_i - \alpha_j)^2\right\}}_{\text{structured ICAR prior}} \\ \times \underbrace{(\tau^2)^{-n/2} \exp\left\{-\frac{1}{2\tau^2} \sum_i \beta_i^2\right\}}_{\text{heterogeneous prior}} \times \underbrace{\pi(\sigma^2) \pi(\tau^2)}_{\Gamma^{-1}(1, 0.01)}$$

- $i \sim j$: contiguous county pair (defined by Queen type adjacency matrix $\mathbf{W}$, i.e. $w_{ij} = 1$ if i, j share a border or vertex, 0 otherwise).
- $\sum_{i \sim j} (\alpha_i - \alpha_j)^2$: **penalises** abrupt spatial jumps.

Why MCMC?

- Poisson $\times$ ICAR $\times$ Normal is **not** conjugate $\Rightarrow$ no closed-form posterior.
- We use **Metropolis-within-Gibbs** with 110,000 iterations (10,000 burn-in).



MCMC: Metropolis-within-Gibbs algorithm

1. Initialise $\Theta^{(0)} = (\alpha, \beta, \theta, \sigma^2, \tau^2)$ arbitrarily; fix random seed.
 2. **For** $t = 1, \dots, 110,000$:
 - **Block 1 – α (ICAR) [MH]**: propose $\alpha_i^* \sim \mathcal{N}(\alpha_i, \delta_\alpha^2)$; accept with $\rho_i = \min(1, p(\alpha_i^*|\cdot)/p(\alpha_i|\cdot))$.
 - **Block 2 – β (heterog.) [MH]**: same random-walk proposal.
 - **Block 3 – θ (covariates) [MH]**: random-walk proposal.
 - **Block 4 – σ^2, τ^2 [Gibbs]**: conjugate $\Gamma^{-1} \Rightarrow$ direct sampling.
 3. Discard first 10,000 draws (burn-in); retain $M = 100,000$.
- Proposal variance δ^2 tuned adaptively to keep acceptance rates in the optimal **20%–65%** window.
 - The ICAR MH ratio combines a Poisson likelihood factor and a spatial penalty factor.
 - Convergence checked via **Geweke Z-test**.



From MCMC to SMRs: posterior inference

Posterior Quantities:

MCMC yields $M = 100,000$ draws of expected deaths: $\tilde{E}_i^{(m)} = E_i e^{\hat{R}_i^{(m)}}$

- **Point estimate (posterior mean):** $SMRs_i = 100 \cdot O_i / \bar{\tilde{E}}_i$
- **90% credible interval** (quantiles inverted since $SMRs \propto 1/\tilde{E}_i$):
 - Lower: $100 \cdot O_i / q_{95\%}(\tilde{E}_i)$
 - Upper: $100 \cdot O_i / q_{5\%}(\tilde{E}_i)$

Posterior Excess-Risk Probability:

$$\Pr(SMRs_i > 100) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\left(\frac{O_i}{\tilde{E}_i^{(m)}} > 1\right)$$

- ≈ 1 : strong evidence of excess risk.
- ≈ 0.5 : ambiguous evidence.
- ≈ 0 : evidence of below-average risk.

Joint interpretation is mandatory: same SMRs can imply very different statistical certainty.

County	SMRs	Pr >100
San Pablo	158.9	0.98
County X	158.9	0.50



Selected results: notable territorial patterns

County	Cancer site	Smoothed SMRs	Pr(SMRs>100)
Montes de Oro	Melanoma/skin	160.4	0.99
Flores	Male genital	310.8	1.00
Turrubares	Breast	180.7	0.99
Dota	Prostate	125.8	0.95
San Mateo	Colon	169.9	0.99

Territorial heterogeneity:

- Substantial variation across 82 counties by tumour site.
- **Chorotega region:** cluster in ill-defined sites and soft tissues.
- **Central Pacific:** pattern in urinary tract and endocrine glands.

Value of Bayesian smoothing:

- Smoothing recovers genuine patterns where the crude SMR would be unreliable.
- San Mateo (15 colon cases): SMRs stabilised using neighbours San Ramón and Esparza.
- Older ages are associated with larger risks overall. Sex matters too.



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Conclusions

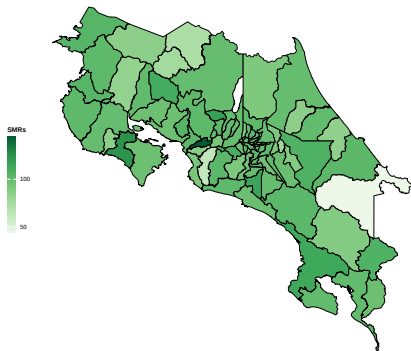
Contributions:

- The model separates **genuine spatial dependence** and resolves the small-area estimation problem.
- $\Pr(\text{SMR}_i > 100)$ provides a **directly interpretable** uncertainty measure for decision-makers.

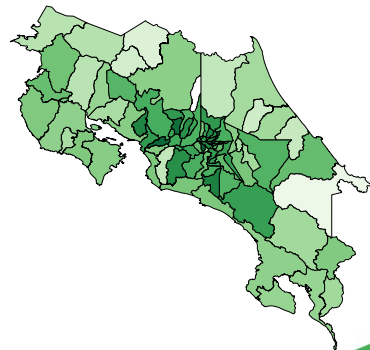
Thank you!

Comments and questions are welcome.

Smooth SMR



Non-smooth SMR





MCMC convergence diagnostics and goodness of fit

- **Geweke test:** compares mean of first 10% vs last 50%; $Z \in [-1.96, 1.96]$ indicates convergence.
- **DIC:** DIC (Deviance Information Criterion) = $\overline{D(\Theta)} + p_D$ (p_D : effective complexity). Lower = better.
- **LMPL (Logarithmic Pseudo-Marginal Likelihood):** sum of conditional predictive log-densities. Higher = better.
- **WAIC (Watanabe-Akaike Information Criterion):** Bayesian counterpart of AIC. Lower = better.
- **Moran Index Test:** evaluates whether there is spacial autocorrelation in the residuals. -1 is perfect dispersion, $+1$ is perfect clustering, 0 indicates a completely random spatial distribution.



Robustness Checks

- **Non-smooth rates:** comparison with non-smooth Standard Mortality Rates. non-smooth rates contain more extreme values and mostly double the values of the smooth rates.
- **Estimation of non-spatial model:** comparison with estimates from a Poisson model without spatial component. Non-spatial estimates suffer from poor fit and non-convergence issues. Their residuals contain spatial autocorrelation.
- **Changes in number of iterations:** models with post burn-in 10,000 or 1,000,000 iterations perform similarly. Best estimation in terms of fit and computational efficiency contains 100,000 iterations.
- **MH acceptance rate:** optimal range **20%–65%**. Large $\delta \rightarrow$ low acceptance; small $\delta \rightarrow$ slow exploration. Not applicable to variances σ^2 and τ^2 .
- **Traceplot:** visual stationarity check post burn-in.



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